

Finding a Percent of a Quantity

Answer Key

Grade 6

Quick Answers

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| 1. 60 | 6. 20 |
| 2. 27 | 7. (a) grams in one square = 4, (b) grams of raisins = 140 |
| 3. (a) ten percent of the seats = 35, (b) thirty percent of the seats = 105 | 8. (a) ten percent of the members = 4.5, (b) sixty percent of the members = 27 |
| 4. (a) 25 percent as a decimal = 0.25, (b) number of blue jackets = 21 | 9. (a) 7.5 percent as a decimal = 0.075, (b) the charge in dollars = 24 |
| 5. (a) ten percent of 60 dollars = 6, (b) five percent of 60 dollars = 3, (c) fifteen percent of 60 dollars = 9 | 10. (a) 20 percent of 150 = 30, (b) 25 percent of 100 = 25, — |
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What is verified

9 of 10 problems fully machine-checked against SymPy.

— marks an answer only you can judge — problem 10. The worked solution states what a correct response must contain.

Curriculum

Level: Grade 6

6.RP.A.3.c – *problems 1, 2, 3, 4, 5, 6, 7, 8, 9, 10*

Difficulty 1–5 of 5 across 19 tagged checks.

- 1.** Hundredths grid: 30 of 100 parts shaded, garden of 200 plants. How many plants?

Solution: Each part of the grid is one hundredth of the garden: $200 \div 100 = 2$ plants per part. Thirty parts are shaded, so $30 \times 2 = 60$ plants. In notation, $\frac{30}{100} \times 200 = 60$. 60

- 2.** Hundredths grid: 45 of 100 parts shaded, shelf of 60 books. How many books?

Solution: One part stands for $60 \div 100 = 0.6$ books, so 45 parts stand for $45 \times 0.6 = 27$ books. In notation, $\frac{45}{100} \times 60 = 27$. 27

- 3.** Tape diagram: 350 seats in 10 strips. Find 10% and 30% of the seats.

Solution: (a) The tape is cut into 10 equal strips, so one strip is $350 \div 10 = 35$ seats. That one strip is 10%. 35

(b) Thirty percent is three of those strips: $3 \times 35 = 105$ seats. Checking with notation, $\frac{30}{100} \times 350 = 105$. 105

4. Rack of 84 jackets, 25% blue. Write the percent as a decimal, then multiply.

Solution: (a) A percent is a number of hundredths, so divide by 100: $25 \div 100 = 0.25$. 0.25

(b) Multiply the decimal by the quantity: $0.25 \times 84 = 21$ blue jackets. 21

5. Pizza night cost 60 dollars. Build 15% from ten-percent strips.

Solution: (a) One ten-percent strip is $60 \div 10 = 6$ dollars. 6

(b) Five percent is half of a ten-percent strip: $6 \div 2 = 3$ dollars. 3

(c) Fifteen percent is one whole strip plus a half strip: $6 + 3 = 9$ dollars. Checking, $\frac{15}{100} \times 60 = 9$. 9

6. School of 250 students, 8% absent. How many students are absent?

Solution: Eight percent as a decimal is $8 \div 100 = 0.08$. Multiplying, $0.08 \times 250 = 20$ students. In fraction form the same computation reads $\frac{8}{100} \times 250 = 20$. 20

7. Hundredths grid for a 400 gram bag of trail mix, 35 parts shaded for raisins.

Solution: (a) The 100 parts share the bag evenly, so one part is $400 \div 100 = 4$ grams. 4

(b) The raisins are 35 parts: $35 \times 4 = 140$ grams. In notation, $\frac{35}{100} \times 400 = 140$. 140

8. Hiking club of 45 members. Find 10% and 60% using ten-percent strips.

Solution: (a) One strip is $45 \div 10 = 4.5$ members. A half member is fine here — the strip is a step on the way, not a head count. 4.5

(b) Sixty percent is six strips: $6 \times 4.5 = 27$ members. Checking, $\frac{60}{100} \times 45 = 27$. 27

9. Repair bill of 320 dollars with a 7.5% service charge.

Solution: (a) Divide the percent by 100 and the decimal point moves two places left: $7.5 \div 100 = 0.075$. 0.075

(b) Multiply: $0.075 \times 320 = 24$ dollars of service charge.

24

10. Maya claims 20% of 150 dollars beats 25% of 100 dollars because 150 is bigger. Compute both, then judge the reasoning.

Solution: (a) $\frac{20}{100} \times 150 = 0.2 \times 150 = 30$ dollars.

30

(b) $\frac{25}{100} \times 100 = 0.25 \times 100 = 25$ dollars.

25

(c) **Instructor-judged.** The conclusion happens to be right — 30 dollars is more than 25 dollars — but the stated reason is not. A percent of a quantity depends on *both* numbers, and a larger starting amount can easily lose to a larger percent: 20% of 150 dollars is 30 dollars, while 25% of 200 dollars would be 50 dollars from a start that is larger again. Full credit needs the comparison of the two computed amounts *and* a statement that comparing only the two starting amounts is not enough to decide.
